

Assignment: Estimating Social Interactions with Peer-of-Peers Instruments

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The Economics of Social Interactions
Summer School at Uniandes, 2026

In class, we simulated and estimated a network peer-effects model in which peers' outcomes enter through a CES social norm and peers' characteristics are used as excluded instruments. This assignment asks you to relax that exclusion restriction. In particular, allow direct peers' characteristics to affect individual outcomes directly, and develop an alternative identification strategy using *peers of peers* as excluded instruments.

1 Model

Let y_i denote the GPA chosen by individual i , and let g_{ij} denote the weight that individual i assigns to peer j . For non-isolated individuals, the social norm faced by individual i is a CES aggregator of peers' outcomes:

$$\tilde{y}_i(\beta, \mathbf{y}) = \left(\sum_j g_{ij} y_j^\beta \right)^{1/\beta}.$$

The outcome equation for non-isolated individuals is

$$y_i = \delta x_i' \gamma + \lambda \tilde{y}_i(\beta, \mathbf{y}) + e_i,$$

For isolated individuals:

$$y_i = x_i' \gamma + \varepsilon_i.$$

2 Extension: Direct Effects of Peers' Characteristics

In class, identification relied on instruments z_i constructed from direct peers' characteristics. This assignment asks you to allow for a violation of the exclusion restriction:

$$\mathbb{E}[z_i \varepsilon_i] \neq 0.$$

This violation arises if direct peers' characteristics affect individual outcomes directly, in addition to affecting outcomes through peers' choices. For example, a student's GPA may be affected not only by peers' GPA, but also directly by peers' background, ability, or study environment. In that case, instruments based on direct peers' characteristics are no longer valid excluded instruments, and the original estimator is inconsistent.

Your task is to develop and motivate an alternative identification strategy using *peers of peers*. The idea is to construct instruments from characteristics of individuals who affect i 's peers, but who are not direct peers of i .

3 Required Output

Students may work in groups of up to three. Each group must submit a `.zip` file containing:

- (i) the modified Python code;
- (ii) a PDF note of at most 6 pages.

The submission must be prepared in a zip folder named according to the following convention:

`FirstName1Surname1_FirstName2Surname2_FirstName3Surname3.zip`.

For example, a group composed of Alejandro Monsalve and Pietro Campa should submit a file named:

`AlejandroMonsalve_PietroCampa.zip`.

The Python code must be executable from the folder and must clearly indicate which parts of the original code were modified. These modifications should be marked with comments explaining what was changed and why.

The PDF note must explain:

1. Why the original estimator is inconsistent when direct peers' characteristics affect individual outcomes directly;
2. Why peers of peers' characteristics are relevant instruments for the endogenous social norm;
3. Under which assumption peers of peers' characteristics are valid excluded instruments;
4. The identifying assumption, illustrated with a small network diagram or example distinguishing direct peers from peers of peers;
5. The estimation algorithm, including construction of a valid peer-of-peers instrument and a GMM estimator that accounts for the nonlinearity of the CES social norm. You may use the same GMM estimator as in class, replacing the original direct-peer instruments with the new peer-of-peer instruments;
6. A table reporting the estimated parameters.

You may modify the data-generating process to facilitate estimation, provided that the modification is clearly described and preserves the main identification problem. You are encouraged to include at least one figure, such as a network illustration, a first-stage relationship, or an objective-function profile, together with a short explanation of its main insight.

4 No Limits on Creativity

Items 1-6 are required, but any extra development is encouraged. Possible extensions include tests for weak instruments, clustered or network-robust standard errors, sensitivity to including direct-peer controls, alternative data-generating processes, comparisons with the original estimator that incorrectly excludes direct peer characteristics, or derivation of the equilibrium conditions of the model enriching the utility function with a direct effect of peers' characteristics on optimal choices.

5 Grading Criteria

We will assess your assignment based on the following criteria:

1. Clarity of the explanation of the identification strategy and the assumptions required for identification. Formal derivations are useful but not required. You should clearly explain the intuition behind the identification strategy and the assumptions that justify it;
2. Clarity of the estimation routine. It is mandatory to list cleanly which part of the code was modified and why, and to explain the estimation routine in the note. You should also clearly explain how the peer-of-peers instruments are constructed;
3. Reproducibility of the results. The code should be executable and produce the same results as reported in the note. You should also provide a clear explanation of how to run the code and reproduce the results;

Should any question arise, feel free to contact the instructors via email at p.campa@rffberlin.com. We will be happy to clarify any doubts and provide guidance on the assignment.